

LEELA SUDHA YELURI

Bengaluru, IN 560002 | +917892430742 | leelasudhayeluri11@gmail.com

Professional Summary

Highly skilled and results-driven Hardware Engineer with 9 years of comprehensive experience across **R&D, Production, and Quality Assurance** within the dynamic **Telecom/Digital Communications sector (Cisco Systems)** and the high-reliability R&D and Production, Space divisions of a leading core electronic product development company specializing in critical **Defence and Rocket applications for BEL, ISRO, DRDO.**

Skills

- Integration Routing and Planning, Cable Routing(Internal, External) & Harness
- Failure Analysis, Debugging Technical Documentation, Team Management, BOM Management
- LRU System Basic testing and Integration plate-wise testing
- PCB Assembling, Components Mounting, Soldering, De soldering, Conformal coating
- SMA Cables Crimping & RF Soldering, Wire stripping & tinning, molex crimping, epoxy methods, Tuning, Re-flow methods
- Verification testing, Connectivity testings, 1 PPS testing, batch files running testing, validation testing, vibration testing

Work History

Assembly Integration & Test Engineer 10/2025 to Current
Astrome Technologies – Bengaluru, India

- Led the electrical integration** of complex satellite subsystems, ensuring 100% adherence to rigorous aerospace quality standards and engineering specifications.
- Executed precision PCB assembly** of spaceflight hardware using specialized tooling and ESD-safe protocols to minimize risk during launch environments.
- Performed expert-level SMD/BGA rework**, including the removal and replacement of fine-pitch components.
- Engineered detailed interconnect diagrams and wiring harness specifications using **WireViz**, ensuring 100% traceability for complex satellite bus-to-payload electrical interfaces.
- Diagnosed and debugged complex hardware failures** at the component level, utilizing schematics to perform successful board-level recoveries.
- Developed and refined Standard Operating Procedures (SOPs)** for assembly, resulting in improvement of integration efficiency for subsequent flight models.
- Managed end-to-end functional testing** of payload systems, identifying and resolving critical anomalies prior to final integration.

- **Conducted Root Cause Analysis (RCA)** on non-conformance reports (NCRs), implementing corrective actions that prevented recurrence in high-reliability space systems.
- **Managed Bill of Materials (BOM)**, identifying and sourcing hardware differences for critical system upgrades and version control.

Debug & Failure Analysis Engineer

12/2024 to 09/2025

Flextronics – Bengaluru, India

- Specialized in component-level debugging and failure analysis of industrial electronic devices, including **Cisco Routers and Switches**
- Deeply Looking to Pull the Customer failure boards from Cisco Clients with help of the team and using **Cisco Agile Methodologies** Website.
- Proactively addressed and resolved complex issues on **Cisco NCS-55A2-MOD, Cisco NCS-57C3-MOD, Modular Devices, Catalyst 9200 Series, Catalyst 9300 Series, Catalyst 9400 Series, Catalyst 9500 Series, Catalyst 9600 Series, Catalyst 1000 Series** by employing Fault Duplication and Isolation Methodologies, Resulting in quantifiable positive outcome, e.g., reduced downtime, Improved network stability.
- Ensure to Collecting the FA Board details Like Onsite facing issues, site installation pictures, Failure logs, Service history & Repair history and shipment related details from **Cesium and QMS Tools**
- Preparing & Maintaining the Lab & Rework Setup for Analysis & Testing purposes
- Proficiently executed a comprehensive suite of diagnostic tests, including **Boot IOS, Ambient DIAG (Diagnostics) Test, 4 corner Traffic Test + 4 corner DIAG Test, 2 corner Traffic Test + 2 corner DIAG Test, OBFL (On Board Failure Log) Tests, Redundancy (Switch over) Test, Regression & Stress Tests**, to effectively Performing, Troubleshooting and Resolve Network Problems.
- Highly Skilled on Rework Process like Removing faulty components replacing (0201, 0402, 0603, 0804) SMD, PTH, QFN, Port and BGA Components On **PCB Motherboards**
- Coordinating & Isolating hardware components, Field Introduced contaminations, mechanical damages, field introducing electrical damages, Missing critical Components related to Issues & sending them to FARC or Bag & Tag Process
- Capturing and Collecting Each and Every Evidence that are related to failure issues through the Test Logs Using **Teraterm, Red hat, Apollo Tools**, Impedance & Voltage Measurements with using multimeters, Clock signals with using Oscilloscope.
- Applied an in-depth understanding of industrial communication protocols like Flash, SPI, JTAG, UART
- Contributed to the testing of management system algorithms, ensuring their accuracy and efficiency in the context of Cisco router functionalities
- Preparing the PPT and Discussing with Cisco Customers for expressing the complete Failure Analysis report in scheduled meetings (QIT)
- Took ownership of functional and performance validation for new features and system enhancements on Cisco Devices, guaranteeing robust and efficient operation.
- Maintaining the LAB equipments with timely calibrations, ESD, 5's Processes

UNISTRING TECH SOLUTIONS – Hyderabad

- Leading the **assembly and integration of Defence and Aerospace components** and systems in compliance with engineering specifications.
- **Led a team of 30 members** towards complete assembly and integration of complex cable harness systems, ensuring all components met stringent design and quality specifications followed by successful system deployments
- Ensuring all assembly and **power-on tests, functional test** activities are conducted following best practices and in adherence to **ISRO & DRDO guidelines**.
- Implemented **EMI/EMC-complaint cable harness and LRU integrations**(shielding solutions, and grounding techniques) systems to mitigate conducted emissions, radiated susceptibility for Defence applications.
- Verify and validate **end-to-end cable harness and integration based on LRU documentation & Final verification testing** by using of DMM, 4 CH CRO, Antennas, power supply setup, clock setup, converters
- Troubleshooting & debugging issues by using different types of techniques like **CAPA, 7 QC tools**.
- Collaborated cross-functionally with engineering and quality assurance teams to resolve design issues and validate system functionality, ensuring compliance with industry standards like **IPC-A-620**.
- Participating in anomaly investigations to determine root causes of failures and propose corrective actions.
- Collaborating with software engineers to integrate **VLSI hardware code into FPGA chips**, performing comprehensive integration testing, and providing GO/NO GO decisions for release.
- Interacting with customers to Resolved complex technical issues for customers onsite, ensuring timely and effective problem resolution and customer satisfaction.

Production technician

08/2016 to 07/2019

CENTUM ELECTRONICS LTD – Bengaluru

- Ensure compliance with **ISO 9001** and **ISO 14001, AS9100, IPC-A-610, 5S** standards, adhering to world-class manufacturing practices and quality standards of **ISRO-PAX-305**
- Experienced in reading and analyzing the documents like **BLUEPRINTS, AST, ASB, Schematics**
- Displayed proficiency with components of various packages in **PCB assembling**(Both SMD and PTH components) with 0201, 0402, 0805, 0603) packages and different QFN IC's
- Hands-on experience with various **soldering, DE-soldering, and rework methods, trouble shooting problems, debugging**
- Proficient in **MMIC** attachments using epoxy methods
- Adhere to **Defence project processes**, including die attachment, S/P substrate to package attachment, wire bonding, and non-conductive and conductive buildup & conformal coating processes

- Implement and maintain clean room standards, including **Class 100, Class 10,000, and Class 100,000** ensuring a controlled and sterile manufacturing environment for **Defence and Aerospace products**

Education

B.Tech: Electronics And Communications Engineering 05/2021

Lenora College of Engineering (JNTU K-University) - Polavaram, India

Diploma: Electronics And Communications Engineering 05/2016

Aditya Engineering College (JNTU K-University) - Kakinada, India

Certifications

ISRO STANDARD OF MICROWAVE INTEGRATED CIRCUITS FABRICATION (ISRO PAX 305) of space application center, Ahmedabad